CSE370 : Database Systems

Assignment 03 | Spring 2026

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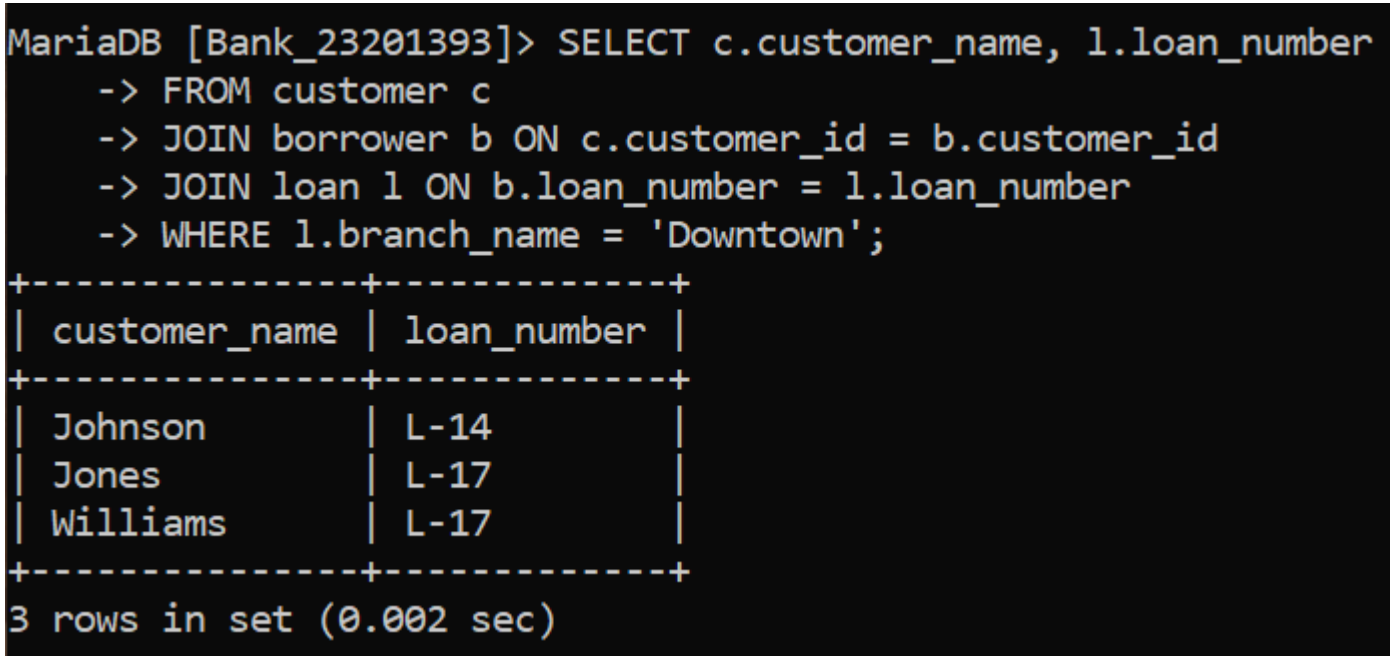
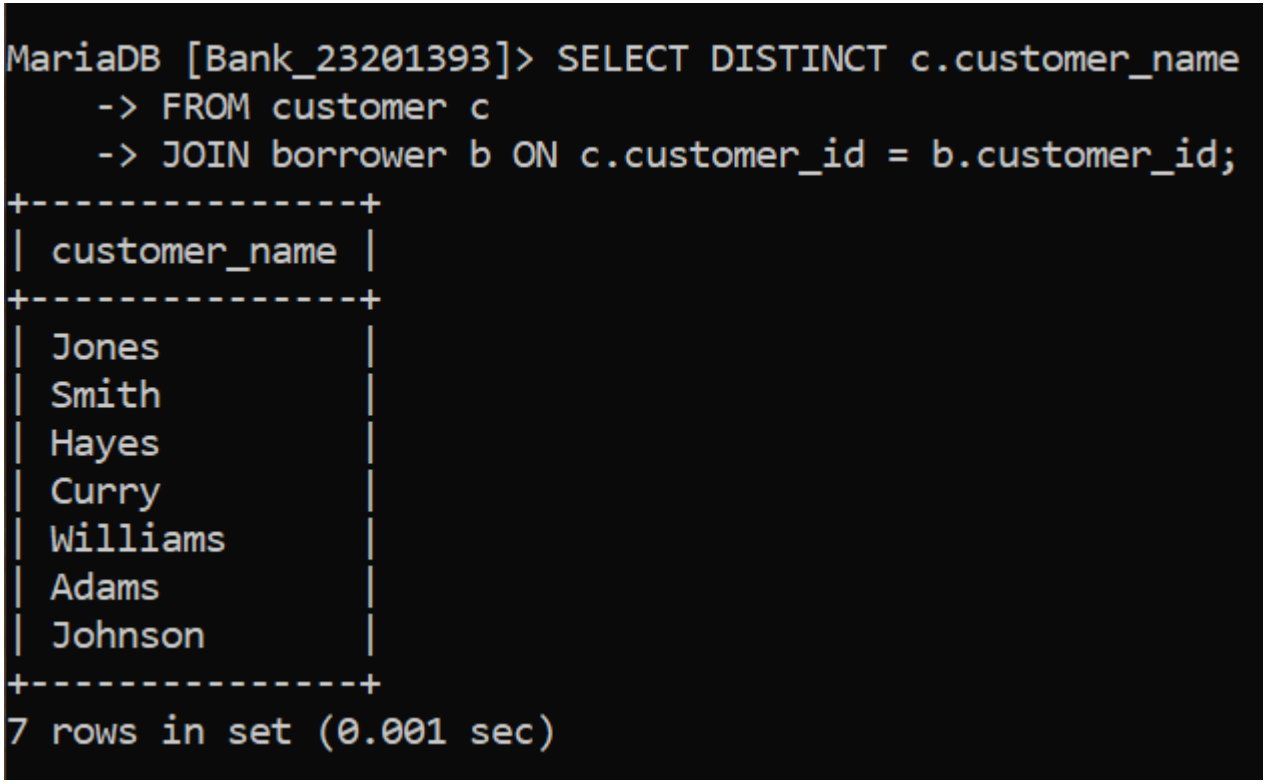
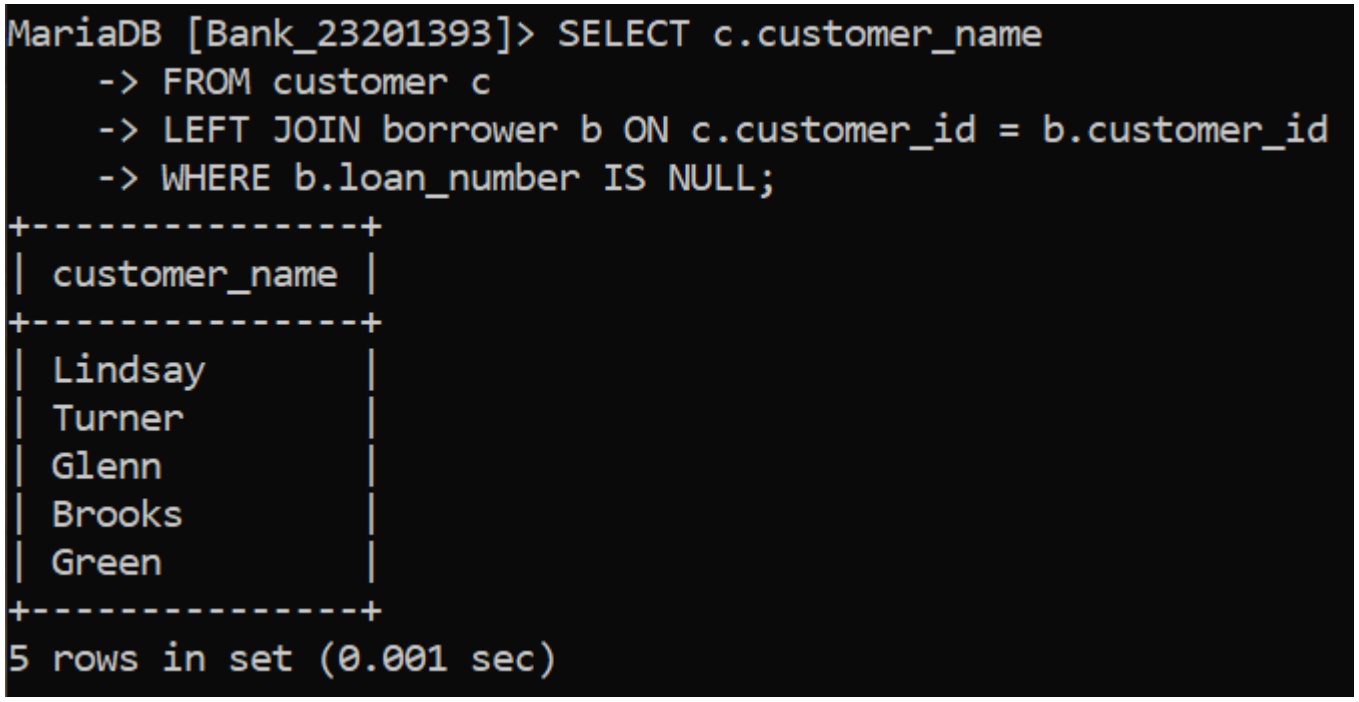
No 1 Query (as Plain Text)	SELECT c.customer_name, l.loan_number FROM customer c JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number WHERE l.branch_name = 'Downtown';
No 1 SS (of Query & Output in Shell)	
No 2 Query (as Plain Text)	SELECT DISTINCT c.customer_name FROM customer c

```
MariaDB [Bank_23201393]> SELECT c.customer_name, l.loan_number
    -> FROM customer c
    -> JOIN borrower b ON c.customer_id = b.customer_id
    -> JOIN loan l ON b.loan_number = l.loan_number
    -> WHERE l.branch_name = 'Downtown';
+---------------+-------------+
| customer_name | loan_number |
+---------------+-------------+
| Johnson       | L-14        |
| Jones         | L-17        |
| Williams      | L-17        |
+---------------+-------------+
3 rows in set (0.002 sec)
```

	JOIN borrower b ON c.customer_id = b.customer_id;
No 2 SS (of Query & Output in Shell)	 MariaDB [Bank_23201393]> SELECT DISTINCT c.customer_name -> FROM customer c -> JOIN borrower b ON c.customer_id = b.customer_id; +----------+ | customer_name | +----------+ | Jones | | Smith | | Hayes | | Curry | | Williams | | Adams | | Johnson | +----------+ 7 rows in set (0.001 sec)
No 3 Query (as Plain Text)	SELECT c.customer_name FROM customer c LEFT JOIN borrower b ON c.customer_id = b.customer_id WHERE b.loan_number IS NULL;

No 3 SS (of Query & Output in Shell)	

```
MariaDB [Bank_23201393]> SELECT c.customer_name
    -> FROM customer c
    -> LEFT JOIN borrower b ON c.customer_id = b.customer_id
    -> WHERE b.loan_number IS NULL;
+---------------+
| customer_name |
+---------------+
| Lindsay       |
| Turner        |
| Glenn         |
| Brooks        |
| Green         |
+---------------+
5 rows in set (0.001 sec)
```

No 4 Query (as Plain Text)	SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.customer_city AS City FROM customer c1 JOIN customer c2 ON c1.customer_city = c2.customer_city AND c1.customer_id < c2.customer_id;

No 4 SS (of Query & Output in Shell)	

```
MariaDB [Bank_23201393]> SELECT c1.customer_name AS Customer1,
    ->  c2.customer_name AS Customer2,
    ->  c1.customer_city AS City
    -> FROM customer c1
    -> JOIN customer c2
    -> ON c1.customer_city = c2.customer_city
    -> AND c1.customer_id < c2.customer_id;
+-----------+-----------+------------+
| Customer1 | Customer2 | City       |
+-----------+-----------+------------+
| Jones     | Hayes     | Harrison   |
| Smith     | Curry     | Rye        |
| Lindsay   | Adams     | Pittsfield |
| Turner    | Green     | Stamford   |
+-----------+-----------+------------+
4 rows in set (0.001 sec)
```

No 5 Query (as Plain Text)	SELECT branch_name, ROUND(SUM(balance * 0.04), 2) AS Total_Interest FROM account GROUP BY branch_name;

No 5 SS
(of Query & Output in Shell)

```
MariaDB [Bank_23201393]> SELECT branch_name,
    ->  ROUND(SUM(balance * 0.04), 2) AS Total_Interest
    -> FROM account
    -> GROUP BY branch_name;
+-------------+----------------+
| branch_name | Total_Interest |
+-------------+----------------+
| Brighton    |          66.00 |
| Downtown    |          20.00 |
| Mianus      |          28.00 |
| Perryridge  |          16.00 |
| Redwood     |          28.00 |
| Round Hill  |          14.00 |
+-------------+----------------+
6 rows in set (0.001 sec)
```

No 6 Query
(as Plain Text)

```
SELECT a.account_number, b.branch_city
FROM account a
JOIN branch b ON a.branch_name = b.branch_name
WHERE a.balance = (
 SELECT MAX(a2.balance)
 FROM account a2
 JOIN branch b2 ON a2.branch_name = b2.branch_name
 WHERE b2.branch_city = b.branch_city
);
```

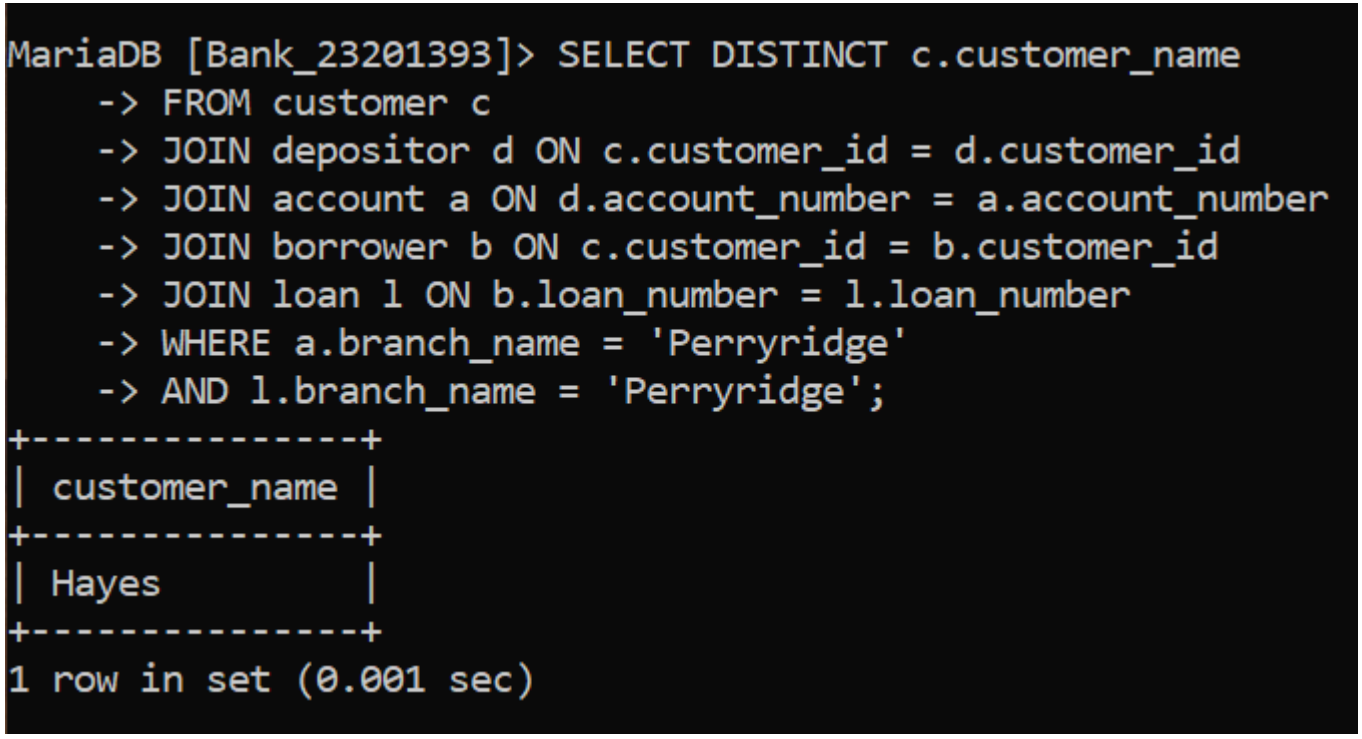
No 6 SS (of Query & Output in Shell)	

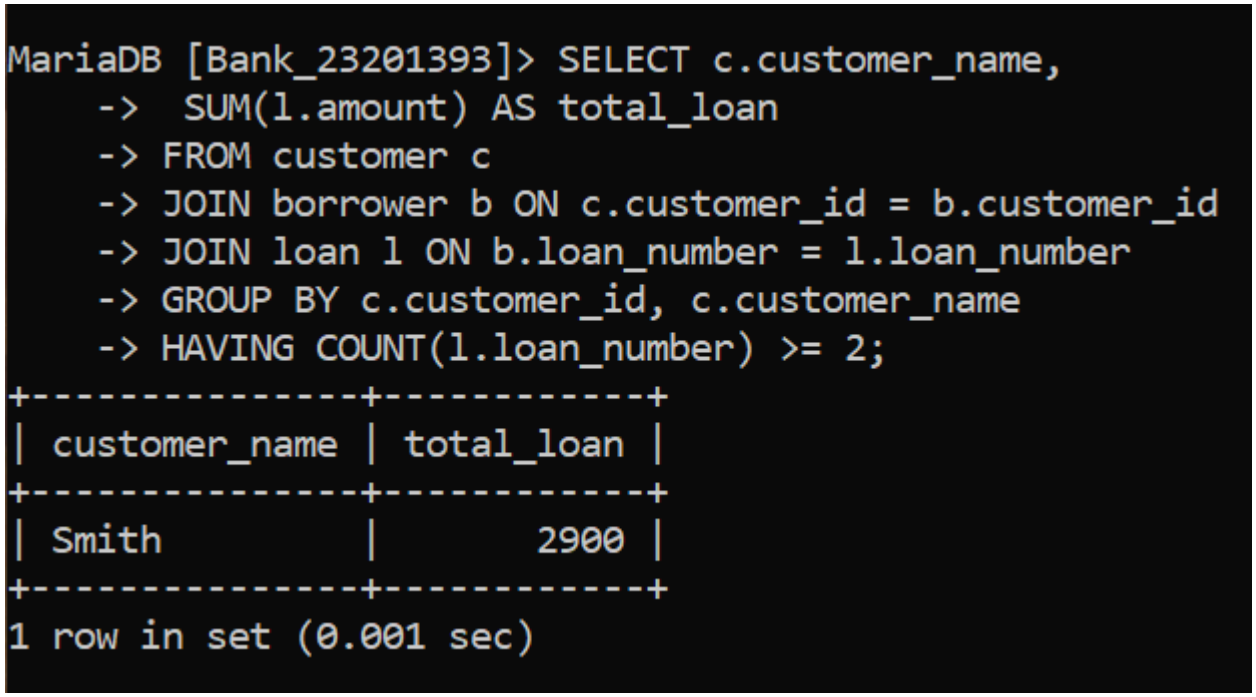
```
MariaDB [Bank_23201393]> SELECT a.account_number, b.branch_city
    -> FROM account a
    -> JOIN branch b ON a.branch_name = b.branch_name
    -> WHERE a.balance = (
    ->  SELECT MAX(a2.balance)
    ->  FROM account a2
    ->  JOIN branch b2 ON a2.branch_name = b2.branch_name
    ->  WHERE b2.branch_city = b.branch_city
    -> );
+----------------+-------------+
| account_number | branch_city |
+----------------+-------------+
| A-201          | Brooklyn    |
| A-215          | Horseneck   |
| A-222          | Palo Alto   |
+----------------+-------------+
3 rows in set (0.001 sec)
```

No 7 Query (as Plain Text)	SELECT loan_number, amount, customer_name FROM (SELECT l.loan_number, l.amount, c.customer_name FROM loan l JOIN borrower b ON l.loan_number = b.loan_number JOIN customer c ON b.customer_id = c.customer_id

	ORDER BY l.amount DESC, l.loan_number DESC LIMIT 5) AS top_loans ORDER BY amount ASC, loan_number DESC;
No 7 SS (of Query & Output in Shell)	

```
MariaDB [Bank_23201393]> SELECT loan_number, amount, customer_name
    -> FROM (
    ->  SELECT l.loan_number, l.amount, c.customer_name
    ->  FROM loan l
    ->  JOIN borrower b ON l.loan_number = b.loan_number
    ->  JOIN customer c ON b.customer_id = c.customer_id
    ->  ORDER BY l.amount DESC, l.loan_number DESC
    ->  LIMIT 5
    -> ) AS top_loans ORDER BY amount ASC,loan_number DESC;
+-------------+--------+---------------+
| loan_number | amount | customer_name |
+-------------+--------+---------------+
| L-17        |   1000 | Jones         |
| L-16        |   1300 | Adams         |
| L-15        |   1500 | Hayes         |
| L-14        |   1500 | Johnson       |
| L-23        |   2000 | Smith         |
+-------------+--------+---------------+
5 rows in set (0.001 sec)
```

No 8 Query (as Plain Text)	SELECT DISTINCT c.customer_name FROM customer c JOIN depositor d ON c.customer_id = d.customer_id JOIN account a ON d.account_number = a.account_number JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number WHERE a.branch_name = 'Perryridge' AND l.branch_name = 'Perryridge';
No 8 SS (of Query & Output in Shell)	
No 9 Query	SELECT c.customer_name,

(as Plain Text)	SUM(l.amount) AS total_loan FROM customer c JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number GROUP BY c.customer_id, c.customer_name HAVING COUNT(l.loan_number) >= 2;
No 9 SS (of Query & Output in Shell)	 MariaDB [Bank_23201393]> SELECT c.customer_name, -> SUM(l.amount) AS total_loan -> FROM customer c -> JOIN borrower b ON c.customer_id = b.customer_id -> JOIN loan l ON b.loan_number = l.loan_number -> GROUP BY c.customer_id, c.customer_name -> HAVING COUNT(l.loan_number) >= 2; +----------+----------+ | customer_name | total_loan | +----------+----------+ | Smith | 2900 | +----------+----------+ 1 row in set (0.001 sec)
No 10 Query (as Plain Text)	SELECT d.customer_id, SUM(a.balance) AS total_balance FROM depositor d JOIN account a ON d.account_number = a.account_number

	GROUP BY d.customer_id HAVING COUNT(d.account_number) > 1;
No 10 SS (of Query & Output in Shell)	

```
MariaDB [Bank_23201393]> SELECT d.customer_id,
    ->  SUM(a.balance) AS total_balance
    -> FROM depositor d
    -> JOIN account a ON d.account_number = a.account_number
    -> GROUP BY d.customer_id
    -> HAVING COUNT(d.account_number) > 1;
+-------------+---------------+
| customer_id | total_balance |
+-------------+---------------+
| C-226       |          1400 |
+-------------+---------------+
1 row in set (0.001 sec)

MariaDB [Bank_23201393]>
```